

1. Administrative & Study Identification

Full Study Title: A Study of Atezolizumab With or Without Tiragolumab in Participants With Unresectable Esophageal Squamous Cell Carcinoma Whose Cancers Have Not Progressed Following Definitive Concurrent Chemoradiotherapy **Short Title/Acronym:** SKYSCRAPER-07 **Protocol Number:** Y042137 **NCT Number:** NCT04543617 **Posting ID:** 439756731334449711 **Sponsor Name:** Hoffmann-La Roche / Roche **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Investigational Product:** Atezolizumab (Trade Name: Tecentriq, ATC Code: L01FF05, anti-PD-L1) and Tiragolumab (anti-TIGIT) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of tiragolumab plus atezolizumab compared with placebo in participants with unresectable locally advanced esophageal squamous cell carcinoma (ESCC) whose cancers have not progressed following definitive concurrent chemoradiotherapy (dCRT), primarily measured by investigator-assessed Progression-Free Survival (PFS). **Secondary Objectives:** To evaluate Overall Survival (OS) across treatment arms, Independent Review Facility (IRF)-assessed PFS, Duration of Response (DoR), and overall safety and tolerability profiles.

2.2 Background & Rationale To address the poor prognosis in locally advanced, unresectable ESCC, where over 50% of patients experience locoregional or distant recurrence after standard-of-care dCRT. Dual immune checkpoint blockade targeting both PD-L1 (atezolizumab) and TIGIT (tiragolumab) is hypothesized to provide synergistic immune activation, serving as an effective consolidation therapy to prevent relapse and prolong survival.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1 (Arm A: Tiragolumab + Atezolizumab | Arm B: Atezolizumab + Placebo | Arm C: Double Placebo)

3.2 Planned Enrollment & Duration Target \$N\$: 760 participants enrolled **Number of Sites:** Global multicenter **Estimated Study Start:** Late 2020 **Estimated Primary Completion:** February 2025 (Primary Data Cutoff)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically confirmed unresectable locally advanced ESCC (Stage II-IVA, selected IVB). 3. Cancers have not progressed following platinum-based definitive concurrent chemoradiotherapy (dCRT). 4. ECOG Performance Status of 0 or 1.

4.2 Exclusion Criteria 1. Prior treatment with CD137 agonists or immune checkpoint blockade therapies (including anti-CTLA-4, anti-PD-1, anti-PD-L1, anti-TIGIT). 2. Unresolved toxicity of CTCAE Grade ≥ 2 from prior chemoradiation therapy. 3. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or evidence of active pneumonitis. 4. Treatment with any other investigational agent, including EGFR inhibitors, with therapeutic intent for esophageal cancer prior to randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A:** Tiragolumab 600 mg + Atezolizumab 1200 mg once every 3 weeks (Q3W) * **Arm B:** Atezolizumab 1200 mg + Placebo Q3W * **Arm C:** Placebo + Placebo Q3W **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Up to 17 cycles of 21 days (approx. 1 year), or until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care per institutional guidelines. **Prohibited:** Concurrent treatment with other investigational agents or immune checkpoint inhibitors during the study period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening: Informed Consent, Medical History, Tumor Assessment, Labs
2	Day 0	Baseline: Randomization, First Dose Administration
3-17	Every 3 Weeks (Q3W)	Interim: Safety Labs, Efficacy/Tumor Assessments (e.g., Q9W scans), IV Infusion
18	Up to Cycle 17 / Progression	End of Study: Final Efficacy Assessment, Safety Follow-up, Transition to Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Investigator-assessed Progression-Free Survival (PFS) comparing Arm A (Tiragolumab +

Atezolizumab) vs. Arm C (Double Placebo). **Measurement Method:** Radiographic tumor assessment evaluated using RECIST v1.1 criteria.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) evaluated hierarchically (Arm A vs C, then Arm B vs C). * Independent Review Facility (IRF)-assessed PFS. * Duration of Response (DoR). * Safety and adverse event (AE) profiles.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard AE reporting per CTCAE guidelines, with targeted monitoring for immune-mediated adverse events (imAEs). **Serious Adverse Events (SAEs):** Required to be reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) established to routinely review safety and efficacy signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** ~760 participants adequately powered for hierarchical statistical testing of PFS and OS. **Handling of Missing Data:** Standard survival analysis methodologies (Kaplan-Meier estimates), utilizing right-censoring at the date of last contact for event-free patients.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standardized onsite and remote clinical monitoring visits ensuring data integrity and protocol compliance. **Audit Trail:** Robust eCRF tracking, comprehensive source data verification, and data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: {{Y042137}} **Posting ID:** {{439756731334449711}}
Registry Number: {{NCT04543617}} **Trial Status:** {{ACTIVE_NOT_RECRUITING}} **Sponsor/Lead Organization:** {{Hoffmann-La Roche / Roche}} **Study Title:** {{SKYSCRAPER-07}} **Official Regulatory Title:** {{A Phase III, Randomized, Double-Blind, Placebo-Controlled Study of Atezolizumab With or Without Tiragolumab (Anti-TIGIT Antibody) in Patients With Unresectable Esophageal Squamous Cell Carcinoma Whose Cancers Have Not Progressed Following Definitive Concurrent Chemoradiotherapy}}

12. Clinical Framework (ESCC Specific)

Primary Condition: {{Esophageal Squamous Cell Carcinoma (ESCC)}}
Preferred Name: {{Metastatic Esophageal Squamous Cell Carcinoma}}
Preferred UMLS Name: {{Unresectable Esophageal Squamous Cell Carcinoma}} **Semantic Type:** {{Neoplastic Process}} **Condition Definition:** {{A squamous cell carcinoma that arises from the esophagus and has metastasized to another anatomic site.}} **MeSH Code:** {{C04.588.322.250}} **SNOMED ID:** {{254636009}} **Diagnosis Threshold:** {{Unresectable Stage II-IVA / IVB without progression post-dCRT}} **Medical Methodology:** {{Consolidation Immune Checkpoint Blockade Therapy}} **EMA Product Info:** {{https://www.ema.europa.eu/en/documents/product-information/tecentriq-epar-product-information_en.pdf}} **Optimization Target:** {{Prevention of locoregional or distant recurrence}}

13. Study Procedures & Methodology

Management Pathway: {{Maintenance/Consolidation pathway following dCRT}} **Education Protocol (HCP):** {{Site initiation and ongoing immune-related AE (irAE) management training}} **Education Protocol (Patient):** {{Informed consent regarding checkpoint inhibitor toxicities and self-reporting criteria}} **Quality Audit Mechanism:** {{Routine Sponsor monitoring and iDMC oversight}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 17 treatment cycles + Long-term survival follow-up}} **Onsite Visit Cadence:** {{Every 3 weeks (Q3W) for treatment dosing}} **Remote Monitoring Frequency:** {{Periodic long-term survival follow-up contacts}} **Checklist Review Interval:** {{Tumor assessments every 9 weeks (Q9W) for the first year, then Q12W}}

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					